MySpot

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Table of Contents

1. Problem Definition.....3
2. System Users and Privileges.....4
3. Entities and Their Descriptions.....5
4. Relationships and Their Descriptions.....6
5. Entity Relationship Diagram (ERD).....7
6. Team Roles and Contributions (Phase II).....8

1. Problem Definition

The proposed system, MySpot, addresses the challenge of managing study-location information in a structured, scalable, and reliable manner for the Virginia Tech and Blacksburg community. Currently, students rely on informal and unstructured sources such as word of mouth, social media posts, messaging groups, and general review platforms to find study spots. These approaches do not provide consistent, searchable, or validated information, leading to fragmented knowledge, duplication, and outdated details.

Existing processes lack a formal data model to represent study environments and their attributes. Information about amenities (e.g. power outlets, Wi-Fi, noise level, seating), availability, operating hours, and room-specific characteristics is often described inconsistently and without standardized terminology. Additionally, there is no structured way to represent hierarchical relationships such as a room belonging to a floor, a floor belonging to a building, and a building belonging to a larger location. Without clearly defined relationships, users cannot efficiently query or compare locations based on precise criteria.

These inefficiencies highlight the need for a centralized, database-driven solution that models real-world entities and their interactions. A structured relational database enables the system to define core entities such as User, Study Spot, Building, Floor, Room, Review, Rating, Amenities, Photos, Favorites, and Reported Content, along with well-defined relationships among them. For example:

- A User may create multiple Reviews, but each Review must reference exactly one Study Spot.
- A Study Spot may have multiple Amenities and Photos.
- A room must belong to one Floor, and each Floor must belong to one Building.
- A User may save multiple Study Spots, but each saved record must reference valid entries in the system.

Through the use of primary keys, foreign keys, and integrity constraints, the database ensures consistency, prevents duplication, and enforces referential integrity across all interactions. Role-based access control further differentiates permissions for Students, Administrators, and Guest users, ensuring that data modification and moderation follow structured workflows.

The scope of the problem extends beyond simply listing locations. The system must support hierarchical location modeling, structured user-generated content, moderation and reporting mechanisms, rating aggregation, and attribute-based filtering. These requirements cannot be reliably implemented without a formal schema that captures entities, relationships, and constraints in a consistent manner.

By defining a clear and normalized data structure that reflects real-world study environments and user interactions, MySpot transforms scattered and unstructured recommendations into a centralized, queryable, and reliable information system.

2. System Users and Privileges

User Type 1: Student At Virginia Tech

Privileges:

- Register and authenticate using a verified VT email address
- Create, edit, and delete study spot entries they have submitted
- Upload and manage photos associated with their submitted study spots
- Submit reviews and ratings for study spots
- Edit or delete their own reviews
- Report inaccurate or inappropriate content
- Save and manage favorite study spots
- View all publicly available study spot information

User Type 2: Platform Administrator

Privileges:

- Create, edit, and delete any study spot entry
- Approve or reject newly submitted study spots
- Edit or remove any user-submitted content (spots, reviews, photos)
- Review and resolve reported content
- Flag content as inappropriate
- Suspend or restrict user accounts
- Access system-wide activity logs and moderation records

User Type 3: Prospective Student/Visitor

Privileges:

- Access the system in read-only mode
- View all publicly available study spot information
- Search and filter study spots based on available attributes

3. Entities and Their Descriptions

Entity 1: user

This entity represents registered users of the system, including Virginia Tech students and platform administrators. It stores identity and authentication-related information and is central to ownership of user-generated content such as study spot submissions, reviews, favorites, and reports.

Entity 2: roles

This entity represents the permission categories in the system (e.g., student, administrator, guest). It supports role-based access control by defining which actions a user is authorized to perform.

Entity 3: spot

This entity represents study locations listed on the platform, such as buildings, floors, rooms, or other study-friendly areas. It stores core study spot information and supports hierarchical organization to model real-world containment (e.g., building → floor → room). Spots are central to reviews, media, amenities, favorites, and reporting.

Entity 4: spot_hours

This entity captures operating hours for a study spot. It supports functionality related to displaying availability by day/time and helps users determine whether a location is open during intended study times.

Entity 5: spot_media

This entity represents media files (primarily photos) uploaded by users and associated with a study spot. It supports richer spot listings and improves user trust by providing visual evidence of the location.

Entity 6: attributes

This entity stores standardized amenity/attribute definitions (e.g., “Power Outlets,” “Quiet,” “Wi-Fi,” “Food Allowed”). It supports consistency across spot listings and enables filter/search functionality based on amenities.

Entity 7: spot_attributes

This entity represents the association between a spot and its amenities/attributes. It allows each spot to be tagged with multiple standardized attributes and optionally stores attribute-specific values (e.g., “High/Medium/Low noise level”), enabling amenity-based querying and filtering.

Entity 8: reviews

This entity represents user-submitted ratings and written feedback for study spots. It captures user experience, supports ranking and reputation, and helps other users make informed decisions about where to study.

Entity 9: favorites

This entity represents a user's saved study spots for quick access. It supports personalization by allowing users to bookmark locations and revisit preferred spots efficiently.

Entity 10: content_report

This entity represents reports submitted by users to flag inaccurate, inappropriate, or misleading content. It supports moderation workflows by tracking report reasons, status, and resolution details for administrative review.

4. Relationships and Their Descriptions

Relationship 1: user - has - roles
Description: This relationship represents assigning a permission level to each user. It exists to enable role-based access control and differentiate privileges across the system.

Relationship 2: user - creates - spot
Description: This relationship represents users submitting new study spot listings. It exists to support community-driven content and enables tracking of ownership of new spots.

Relationship 3: spot - has - spot_hours
Description: This relationship represents the operating hours associated with a study spot. It exists so users can determine when a location is open and available and enables time-based availability functionality.

Relationship 4: spot - has - spot_media
Description: This relationship represents photos attached to a study spot. It exists to provide visual information about study environments and enhance listing credibility.

Relationship 5: user - uploads - spot_media
Description: This relationship represents the user who uploaded a photo. It exists to track ownership and enables accountability and content moderation functionality.

Relationship 6: user - is described by - attributes
Description: This relationship represents the amenities available at a study spot. It exists to support filtering, comparison, and attribute-based querying functionality based on study preferences.

Relationship 7: user - writes - reviews
Description: This relationship represents users creating reviews and ratings. It exists to capture user experiences and enables feedback-driven evaluation of study spots.

Relationship 8: reviews - refers to - spot
Description: This relationship represents reviews being attached to a specific study spot. It exists so users can evaluate locations based on feedback and enables reputation scoring functionality.

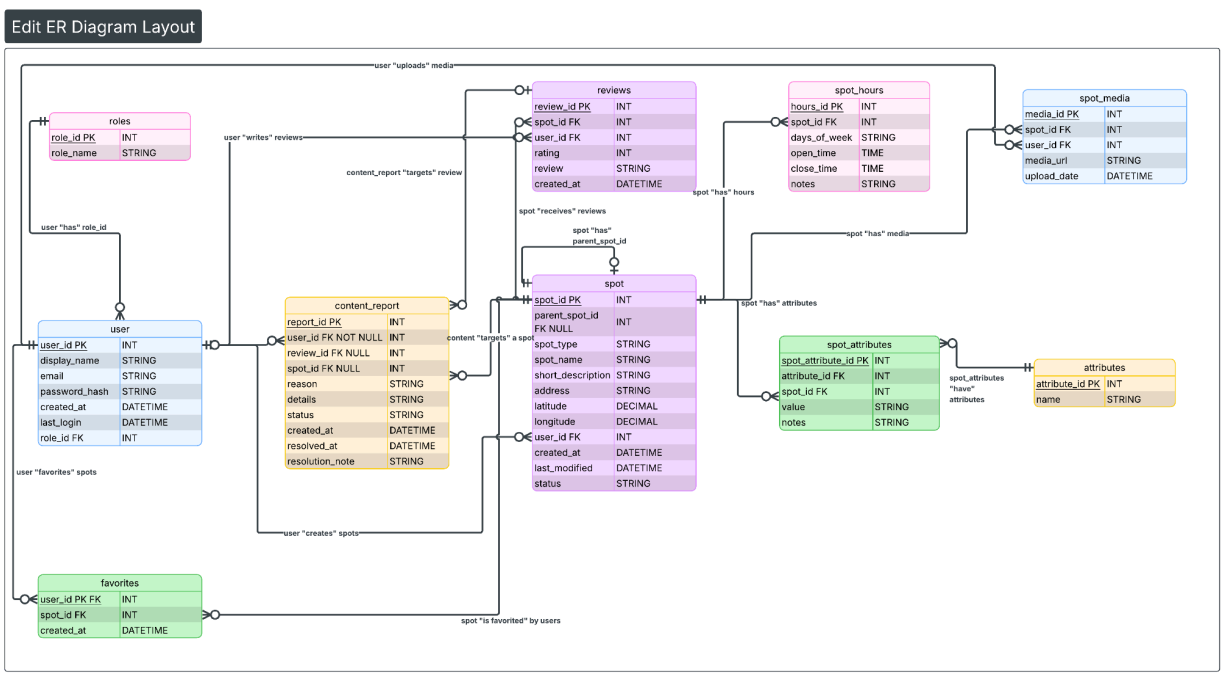
Relationship 9: user - saves - favorites
Description: This relationship represents users bookmarking study spots. It exists to support personalized lists of preferred locations and enables quick-access functionality.

Relationship 10: user - submits - content_report
Description: This relationship represents users reporting inappropriate or incorrect content. It exists to support moderation workflows and platform safety functionality.

Relationship 11: content_report - targets - spot
Description: This relationship represents reports about study spot listings. It exists to allow administrators to review and resolve spot-related issues and maintain data integrity.

Relationship 12: content_report - targets - reviews
Description: This relationship represents reports about specific reviews. It exists to allow administrators to moderate user feedback and enforce community guidelines.

5. Entity Relationship Diagram (ERD)



To access the enlarged version of the Entity Relationship Diagram, please click on the following link:
https://lucid.app/lucidchart/240d7658-5b68-4513-a6aa-ce425d5289d8/edit?viewport_loc=-935%2C-6865%2C3989%2C2061%2C0_0&invitationId=inv_64a8dfad-4f0b-4237-b776-b0c34313380d

6. Team Roles and Contributions (Phase II)

Team Member 1: Lakshya Jain
Role and Contributions:

- Brainstormed entities, their attributes, and the relationships.
- Completed the Problem Definition section
- Proofreading Document

Team Member 2: Jenna Baker
Role and Contributions:

- Brainstormed user privileges
- Updated and finalized System Users and Privileges section
- Brainstormed entity connections and relationships
- Completed Relationships and Their Descriptions sections
- Reviewed document and fixed any flaws

Team Member 3: Liam Erickson
Role and Contributions:

- Helped brainstorm entities and their attributes
- Met with the team to form relationships between entities
- Created ER Diagram for the phase

Team Member 4: Tia Mehta
Role and Contributions:

- Brainstormed and added a few user privileges for section 2
- Met with team to discuss entities and their relationships
- Completed Section 3 of the report

AI Team Member: Generative AI (Chatty)
Role and Contributions:

- Helped give ideas on entity ideas
- Helped connect relationships between entities
- Created the backbone of the ERD